

Result:
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Neptun ID:
Name: _____
Draft papers: ☐

The test consists of **3 questions**. The available time to solve the quiz is **20 minutes**. Please use a blue coloured pen; the use of any other tools is forbidden.

1.) (essay)

	a	b	c	d
2.)	☐	☐	☐	☐
3.)	☐	☐	☐	☐

Please copy the multiple choice questions' solution in the attached table. For each correct answer marked, 2 points are awarded; for each incorrect answer marked, -1 point is awarded. There is no limit to the number of correct answers per question. A negative aggregate score will be interpreted as 0 points.

1.) Define the semantic value of $(A \wedge B)$, e.g., by using a truth-table!

Score: 2 / 2.

2.) Which two interpretations are models of the following set of formulae?

$$\{ (q \equiv ((q \equiv p) \vee q)), (p \supset \neg q) \}$$

(a) $\varrho(p) = 0, \quad \varrho(q) = 0$

2 $\varrho(p) = 0, \quad \varrho(q) = 1$

2 $\varrho(p) = 1, \quad \varrho(q) = 0$

(d) $\varrho(p) = 1, \quad \varrho(q) = 1$

3.) Which two statements are true regarding the following formula?

$$\neg(q \wedge \neg p)$$

- 2** a disjunctive normal form for the formula above is: $\neg q \vee p$
- (b) a disjunctive normal form for the formula above is: $\neg q \wedge p$
- 2** a conjunctive normal form for the formula above is: $\neg q \vee p$
- (d) a normal form for the formula above is: $\neg p \supset \neg q$

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